

Тема: Настройка протоколов динамической маршрутизации RIP v2 и OSPF

1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

Сеть PC1 (192.168.10.0): 192.168.10.10 (ip) /24 (маска) 192.168.10.1 (шлюз)

Сеть PC2 (192.168.20.0): 192.168.20.10 (ip) /24 (маска) 192.168.20.1 (шлюз)

Сеть PC3 (192.168.30.0): 192.168.30.10 (ip) /24 (маска) 192.168.30.1 (шлюз)

Сеть PC4 (192.168.40.0): 192.168.40.10 (ip) /24 (маска) 192.168.40.1 (шлюз)

Сеть PC5 (192.168.50.0): 192.168.50.10 (ip) /24 (маска) 192.168.50.1 (шлюз)

Сеть свича (10.0.123.0): /24 (маска) R1 (.1), R2 (.2), R3 (.3)

R1-R5 (10.0.15.0): /30 (маска) R5 (.1), R1 (.2)

R4-R5 (10.0.45.0): /30 (маска) R4 (.1), R5 (.2)

R2-R6 (10.0.26.0): /30 (маска) R2 (.1), R6 (.2)

R3-R7 (10.0.37.0): /30 (маска) R3 (.1), R7 (.2)

R3-R8 (10.0.38.0): /30 (маска) R3 (.1), R8 (.2)

R6-R7 (10.0.67.0): /30 (маска) R6 (.1), R7 (.2)

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

настройка R4 роутера:

```
conf t
```

```
int FastEthernet1/0
```

```
ip add 192.168.10.1 255.255.255.0
```

```
no shut
```

```
int FastEthernet0/0
```

```
ip add 10.0.45.1 255.255.255.252
no shut
exit
router rip
version 2
no auto-summary
network 192.168.10.0
network 10.0.0.0
exit
```

настройка R5 роутера:

```
conf t
int Ethernet2/0
ip add 192.168.20.1 255.255.255.0
no shut
int FastEthernet1/0
ip add 10.0.45.2 255.255.255.252
no shut
int FastEthernet0/0
ip add 10.0.15.1 255.255.255.252
no shut
exit
router rip
version 2
no auto-summary
network 192.168.20.0
network 10.0.0.0
exit
```

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2. Зону 1 настроить как полностью (nb!) тупиковую.

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

настройка R1 роутера:

```
conf t
int FastEthernet1/0
ip add 10.0.15.2 255.255.255.252
no shut
int FastEthernet0/0
ip add 10.0.123.1 255.255.255.0
no shut
exit
! Настройка RIP и редистрибуции из OSPF
router rip
version 2
no auto-summary
network 10.0.0.0
redistribute ospf 1 metric 3
exit
router ospf 1
network 10.0.123.0 0.0.0.255 area 0
redistribute rip subnets
exit
```

В терминале R2

```
conf t
int FastEthernet0/0
ip add 10.0.123.2 255.255.255.0
no shut
```

```
int FastEthernet1/0
ip add 10.0.26.1 255.255.255.252
no shut
exit
router ospf 1
network 10.0.123.0 0.0.0.255 area 0
network 10.0.26.0 0.0.0.3 area 2
exit
```

В терминале R3

```
conf t
int FastEthernet0/0
ip add 10.0.123.3 255.255.255.0
no shut
int Ethernet2/0
ip add 10.0.38.1 255.255.255.252
no shut
int FastEthernet1/0
ip add 10.0.37.1 255.255.255.252
no shut
exit
router ospf 1
network 10.0.123.0 0.0.0.255 area 0
network 10.0.37.0 0.0.0.3 area 2
network 10.0.38.0 0.0.0.3 area 1
area 1 stub no-summary
exit
```

Настройка R6:

```
conf t
int Ethernet2/0
ip add 192.168.30.1 255.255.255.0
no shut
int FastEthernet0/0
ip add 10.0.26.2 255.255.255.252
no shut
int FastEthernet1/0
ip add 10.0.67.1 255.255.255.252
no shut
exit
router ospf 1
network 192.168.30.0 0.0.0.255 area 2
network 10.0.26.0 0.0.0.3 area 2
network 10.0.67.0 0.0.0.3 area 2
exit
```

Настройка R7:

```
conf t
int Ethernet2/0
ip add 192.168.40.1 255.255.255.0
no shut
int FastEthernet0/0
ip add 10.0.37.2 255.255.255.252
no shut
int FastEthernet1/0
ip add 10.0.67.2 255.255.255.252
no shut
```

```
exit
```

```
router ospf 1
```

```
network 192.168.40.0 0.0.0.255 area 2
```

```
network 10.0.37.0 0.0.0.3 area 2
```

```
network 10.0.67.0 0.0.0.3 area 2
```

```
exit
```

Настройка R8:

```
conf t
```

```
int FastEthernet1/0
```

```
ip add 192.168.50.1 255.255.255.0
```

```
no shut
```

```
int FastEthernet0/0
```

```
ip add 10.0.38.2 255.255.255.252
```

```
no shut
```

```
exit
```

```
router ospf 1
```

```
network 192.168.50.0 0.0.0.255 area 1
```

```
network 10.0.38.0 0.0.0.3 area 1
```

```
area 1 stub
```

```
exit
```

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

PC1 to PC5

```
[PC1> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=59 time=69.007 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=59 time=71.597 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=59 time=81.806 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=59 time=73.399 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=59 time=92.052 ms
```

PC5 to PC1

```
[PC5> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=59 time=89.167 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=59 time=71.877 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=59 time=96.536 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=59 time=83.692 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=59 time=67.185 ms
```

PC1 to PC2

```
[PC1> ping 192.168.20.10

84 bytes from 192.168.20.10 icmp_seq=1 ttl=62 time=30.975 ms
84 bytes from 192.168.20.10 icmp_seq=2 ttl=62 time=33.419 ms
84 bytes from 192.168.20.10 icmp_seq=3 ttl=62 time=25.639 ms
84 bytes from 192.168.20.10 icmp_seq=4 ttl=62 time=22.682 ms
84 bytes from 192.168.20.10 icmp_seq=5 ttl=62 time=35.843 ms
```

PC2 to PC1

```
[PC2> ping 192.168.10.10

84 bytes from 192.168.10.10 icmp_seq=1 ttl=62 time=47.525 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=62 time=27.765 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=62 time=22.088 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=62 time=31.829 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=62 time=33.699 ms
```

PC1 to PC3

```
[PC1> ping 192.168.30.10
```

```
84 bytes from 192.168.30.10 icmp_seq=1 ttl=59 time=98.465 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=59 time=72.972 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=59 time=74.458 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=59 time=73.920 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=59 time=88.183 ms
```

PC3 to PC1

```
PC3> ping 192.168.10.10
```

```
84 bytes from 192.168.10.10 icmp_seq=1 ttl=59 time=92.346 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=59 time=67.994 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=59 time=68.026 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=59 time=54.857 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=59 time=75.532 ms
```

PC1 to PC4

```
[PC1> ping 192.168.40.10
```

```
84 bytes from 192.168.40.10 icmp_seq=1 ttl=59 time=91.630 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=59 time=85.968 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=59 time=92.507 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=59 time=85.473 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=59 time=73.061 ms
```

PC4 to PC1

```
[PC4> ping 192.168.10.10
```

```
84 bytes from 192.168.10.10 icmp_seq=1 ttl=59 time=81.504 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=59 time=94.936 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=59 time=65.037 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=59 time=85.551 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=59 time=104.928 ms
```

PC2 to PC3

```
[PC2> ping 192.168.30.10
```

```
84 bytes from 192.168.30.10 icmp_seq=1 ttl=60 time=64.836 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=60 time=61.328 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=60 time=72.636 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=60 time=77.026 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=60 time=81.787 ms
```

PC3 to PC2

```
[PC3> ping 192.168.20.10
```

```
84 bytes from 192.168.20.10 icmp_seq=1 ttl=60 time=79.494 ms
84 bytes from 192.168.20.10 icmp_seq=2 ttl=60 time=71.318 ms
84 bytes from 192.168.20.10 icmp_seq=3 ttl=60 time=72.629 ms
84 bytes from 192.168.20.10 icmp_seq=4 ttl=60 time=59.070 ms
84 bytes from 192.168.20.10 icmp_seq=5 ttl=60 time=53.597 ms
```

PC2 to PC4

```
[PC2> ping 192.168.40.10
```

```
84 bytes from 192.168.40.10 icmp_seq=1 ttl=60 time=55.018 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=60 time=63.819 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=60 time=43.923 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=60 time=51.884 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=60 time=53.367 ms
```

```
PC2> █
```

PC4 to PC2

```
[PC4> ping 192.168.20.10
```

```
84 bytes from 192.168.20.10 icmp_seq=1 ttl=60 time=72.747 ms
84 bytes from 192.168.20.10 icmp_seq=2 ttl=60 time=59.953 ms
84 bytes from 192.168.20.10 icmp_seq=3 ttl=60 time=78.873 ms
84 bytes from 192.168.20.10 icmp_seq=4 ttl=60 time=66.462 ms
84 bytes from 192.168.20.10 icmp_seq=5 ttl=60 time=55.821 ms
```

PC2 to PC5

```
[PC2> ping 192.168.50.10
```

```
84 bytes from 192.168.50.10 icmp_seq=1 ttl=60 time=88.528 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=60 time=62.613 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=60 time=51.847 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=60 time=54.377 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=60 time=82.847 ms
```

PC5 to PC2

```
[PC5> ping 192.168.20.10
```

```
84 bytes from 192.168.20.10 icmp_seq=1 ttl=60 time=76.766 ms
84 bytes from 192.168.20.10 icmp_seq=2 ttl=60 time=63.316 ms
84 bytes from 192.168.20.10 icmp_seq=3 ttl=60 time=53.215 ms
84 bytes from 192.168.20.10 icmp_seq=4 ttl=60 time=66.611 ms
84 bytes from 192.168.20.10 icmp_seq=5 ttl=60 time=70.272 ms
```

PC3 to PC4

```
[PC3> ping 192.168.40.10
```

```
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=41.315 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=26.975 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=40.754 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=32.451 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=41.208 ms
```

PC4 to PC3

```
[PC4> ping 192.168.30.10
```

```
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=35.450 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=23.136 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=34.355 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=34.613 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=41.126 ms
```

PC3 to PC5

```
[PC3> ping 192.168.50.10
```

```
84 bytes from 192.168.50.10 icmp_seq=1 ttl=60 time=49.440 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=60 time=54.505 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=60 time=43.762 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=60 time=59.829 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=60 time=52.377 ms
```

PC5 to PC3

```
[PC5> ping 192.168.30.10
```

```
84 bytes from 192.168.30.10 icmp_seq=1 ttl=60 time=82.554 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=60 time=61.712 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=60 time=63.058 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=60 time=63.877 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=60 time=42.803 ms
```

PC4 to PC5

```
[PC4> ping 192.168.50.10
```

```
84 bytes from 192.168.50.10 icmp_seq=1 ttl=61 time=31.032 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=61 time=52.445 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=61 time=44.486 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=61 time=52.815 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=61 time=47.035 ms
```

PC5 to PC4

```
[PC5> ping 192.168.40.10
```

```
84 bytes from 192.168.40.10 icmp_seq=1 ttl=61 time=71.228 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=61 time=56.263 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=61 time=45.257 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=61 time=44.761 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=61 time=56.388 ms
```

6) Перехватить в Wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

Выполняем захват на линке R4-R5

Видим кучу пакетов Response. Тип - 2. Адрес назначения - мультикаст (224.0.0.9). Содержит таблицы маршрутизации с адресами, масками.

```

> Frame 4: Packet, 246 bytes on wire (1968 bits), 246 bytes captured (1968 bits) on interface -, id 0
> Ethernet II, Src: cc:05:06:2e:00:10 (cc:05:06:2e:00:10), Dst: IPv4mcast_09 (01:00:5e:00:00:09)
> Internet Protocol Version 4, Src: 10.0.45.2, Dst: 224.0.0.9
> User Datagram Protocol, Src Port: 520, Dst Port: 520
< Routing Information Protocol
  Command: Response (2)
  Version: RIPv2 (2)
  > IP Address: 10.0.15.0, Metric: 1
  > IP Address: 10.0.26.0, Metric: 4
  > IP Address: 10.0.37.0, Metric: 4
  > IP Address: 10.0.38.0, Metric: 4
  > IP Address: 10.0.67.0, Metric: 4
  > IP Address: 10.0.123.0, Metric: 2
  > IP Address: 192.168.20.0, Metric: 1
  > IP Address: 192.168.30.0, Metric: 4
  > IP Address: 192.168.40.0, Metric: 4
  > IP Address: 192.168.50.0, Metric: 4

```

Выполняем захват на линке R1-свич.

Получаем hello пакеты протокола OSPF. Тип - 1. Содержание ниже. Адрес назначения мультикаст (224.0.0.5)

No.	Time	Source	Destination	Protocol	Length	Info
6	7.732835	10.0.123.1	224.0.0.6	OSPF	98	LS Update
7	7.753273	10.0.123.3	224.0.0.5	OSPF	98	LS Update
8	7.805437	10.0.123.2	224.0.0.5	OSPF	98	Hello Packet
9	8.911533	10.0.123.1	224.0.0.5	OSPF	98	Hello Packet
10	9.060913	10.0.123.3	224.0.0.5	OSPF	98	Hello Packet
11	10.270699	10.0.123.2	224.0.0.5	OSPF	78	LS Acknowledge
18	17.785224	10.0.123.2	224.0.0.5	OSPF	98	Hello Packet
21	18.912194	10.0.123.1	224.0.0.5	OSPF	98	Hello Packet
22	19.070170	10.0.123.3	224.0.0.5	OSPF	98	Hello Packet
23	20.468994	10.0.123.3	224.0.0.5	OSPF	98	LS Update
24	20.518493	10.0.123.3	224.0.0.5	OSPF	98	LS Update

```

Frame 22: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: cc:03:06:1c:00:00 (cc:03:06:1c:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
Internet Protocol Version 4, Src: 10.0.123.3, Dst: 224.0.0.5
Open Shortest Path First
< OSPF Header
  Version: 2
  Message Type: Hello Packet (1)
  Packet Length: 52
  Source OSPF Router: 10.0.123.3
  Area ID: 0.0.0.0 (Backbone)
  Checksum: 0x5389 [correct]
  Instance ID: Base IPv4 Unicast Instance (0)
  Auth Type: Null (0)
  Auth Data (none): 0000000000000000
  > OSPF Hello Packet
  > OSPF LLS Data Block

```

Пакет LS Update. Тип - 4. Содержит LSA, список сетей, их состояний и масок. Информировывает остальную сеть.

```
> Frame 23: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
> Ethernet II, Src: cc:03:06:1c:00:00 (cc:03:06:1c:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
> Internet Protocol Version 4, Src: 10.0.123.3, Dst: 224.0.0.5
> Open Shortest Path First
  > OSPF Header
    Version: 2
    Message Type: LS Update (4)
    Packet Length: 64
    Source OSPF Router: 10.0.123.3
    Area ID: 0.0.0.0 (Backbone)
    Checksum: 0xf737 [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  > LS Update Packet
```

7) Сохранить в отдельные файлы с префиксом `rt_` и именем маршрутизатора таблицы маршрутизации всех маршрутизаторов.

Таблицы маршрутизации сохранены в папке `rt`.

8) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.

Конфиги в папке `conf`.